Proposed concept for middleware operations

AI Incident Copilot for Airport Middleware

From incident detection to intelligent investigation and recommendation

Alert
Incident detected

→

AI Analysis
Correlate and recommend

→

Engineer
Validate and act

AI recommends. The engineer decides. No autonomous production changes.

The Current Incident Investigation Process

Today the engineer stitches this together by hand:

1. **Incident alert**
2. **Engineer starts investigation**
3. **Check MQ**
4. **Check ACE**
5. **Check logs**
6. **Review previous incidents**
7. **Identify pattern**
8. **Decide what to do**

The required information is available, but engineers must manually connect the dots across multiple sources.

AIP Sentinel (existing)

Detection and alerting.

Tells the team that something is wrong so investigation can start.

AI Incident Copilot (proposed addition)

Investigation intelligence layer.

Does not replace Sentinel. Prepares the investigation so the engineer can decide faster.

AI Incident Copilot

Inputs the Copilot would use

- Incident alert
- IBM MQ data
- IBM ACE logs
- Application logs
- Historical incidents
- Knowledge / runbooks

AI Incident Copilot — proposed intelligence layer

1. Analyze
2. Correlate
3. Find patterns
4. Recommend

Engineer validation

Review findings, confirm likely cause, choose the action

Action / resolution

Engineer executes the change — not the AI

The Copilot does not replace the engineer.
It reduces investigation effort and helps the engineer reach the right decision faster.

Example: AODB Incident Investigation

Illustrative path (generic example)

AODB → FLTU / RNUP message → IBM MQ → IBM ACE → Processing error

What the Copilot could produce for the engineer

Incident	Correlation	Pattern	Recommendation
AODB FLTU/RNUP processing failure	Message reached MQ successfully. Failure occurred during downstream processing.	Similar incidents were previously associated with the ACE processing layer.	•Check ACE application logs •Validate message payload •Check affected integration flow •Compare with similar historical incidents

AIP Sentinel

"Something is wrong."
Detection and alerting — the starting point.

AI Incident Copilot

"What happened, what is related, what pattern do we see, and what should the engineer investigate next?"

Future Vision

A staged path — engineer remains in control at every phase

PHASE 1

Assist

AI analyzes incidents and recommends next steps.

PHASE 2

Learn

Use engineer feedback and resolved incidents to improve recommendations.

PHASE 3

Predict

Identify recurring patterns and possible risks earlier.

PHASE 4

Controlled automation

Potential automation only for approved, low-risk actions, with proper validation.

Key benefit

Reduce incident investigation time while keeping the engineer in control.

Thank You

Questions and discussion welcome